

MAT1575, Classwork3, Fall2026

1.2 The Definite Integral (p.24-36)

1. Review: Let f be defined on a closed interval $[a, b]$ and $P = \{x_0, x_1, x_2, \dots, x_n\}$ be a regular partition of $[a, b]$. Let Δx be the width of each subinterval. Then $\Delta x = \frac{b-a}{n}$, $L_n = \sum_{i=1}^n f(x_{i-1}) \Delta x$, and $R_n = \sum_{i=1}^n f(x_i) \Delta x$.
2. Forming Riemann Sums: Following the assumption in 1., let x_i^* be any point in $[x_{i-1}, x_i]$.

A Riemann Sum is defined for $f(x)$ as $\sum_{i=1}^n f(x_i^*) \cdot \Delta x$

Then, the area under the curve $y = f(x)$ on $[a, b]$ is given by $A = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$

3. Upper Sum and Lower Sum: Depending on the choices of x_i^* in $[x_{i-1}, x_i]$, we have, for $i = 1, 2, \dots, n$,

(a) The Riemann Sum is called upper sum if $f(x_i^*) \geq f(x)$ which is the maximum value of $f(x)$ for $x \in [x_{i-1}, x_i]$.

(b) The Riemann Sum is called lower sum if $f(x_i^*) \leq f(x)$ which is the minimum value of $f(x)$ for $x \in [x_{i-1}, x_i]$.

Then, we have lower sum $\leq$ Area of $A \leq$ Upper sum.

4. Definition: Definite Integral.

If $f(x)$ is a function defined on an interval $[a, b]$, the definite integral of f from a to b is given by

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$$

provided the limit exists. If this limit exists, $f(x)$ is said to be integrable on $[a, b]$, or is an integrable function.

The numbers a and b are x -values and are called limits of integration; a is the lower limit and b is the upper limit. Function $f(x)$ is called the integrand, and the dx indicates that is a function with respect to x , called the variable of integration.

5. What is the difference between definite integral and indefinite integral?

A definite integral is a number. An indefinite integral is a family of functions.

6. Example: Evaluating an Integral Using the Definition.

- (a) Use the definition of the definite integral to evaluate $\int_0^8 (x+2) dx$. Use a right endpoint approximation to generate the Riemann sum. $= 48$

$f(x) = x+2$, $I = [0, 8]$, $\Delta x = \frac{8}{n}$, $P = \{0, 0+\Delta x, 0+2\Delta x, 0+3\Delta x, \dots, 0+(n-1)\Delta x, 8\}$ and $x_i = 0 + i \cdot \Delta x$

$$R_n = \sum_{i=1}^n f(x_i) \Delta x = \sum_{i=1}^n f\left(\frac{8}{n} \cdot i\right) \frac{8}{n} = \sum_{i=1}^n \left(\frac{8}{n} \cdot i + 2\right) \frac{8}{n} = \sum_{i=1}^n \left(\frac{64}{n^2} \cdot i + \frac{16}{n}\right) = \sum_{i=1}^n \frac{64}{n^2} i + \sum_{i=1}^n \frac{16}{n}$$

$$\int_0^8 (x+2) dx = \lim_{n \rightarrow \infty} R_n = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x = \lim_{n \rightarrow \infty} \left(48 + \frac{32}{n}\right) = 48$$

- (b) Use the definition of the definite integral to evaluate $\int_{-1}^3 (2x-1) dx$. Use a right endpoint approximation to generate the Riemann sum. $= 4$

$f(x) = 2x-1$, $I = [-1, 3]$, $\Delta x = \frac{4}{n}$, $P = \{-1, -1+\Delta x, -1+2\Delta x, -1+3\Delta x, \dots, -1+(n-1)\Delta x, 3\}$ and $x_i = -1 + i \cdot \frac{4}{n}$

$$R_n = \sum_{i=1}^n f(x_i) \Delta x = \sum_{i=1}^n \left[2\left(-1 + \frac{4}{n}i\right) - 1\right] \cdot \frac{4}{n} = \sum_{i=1}^n \left(\frac{8}{n}i - \frac{4}{n}\right) = \sum_{i=1}^n \frac{8}{n}i - \sum_{i=1}^n \frac{4}{n}$$

$$\int_{-1}^3 (2x-1) dx = \lim_{n \rightarrow \infty} \left(4 + \frac{16}{n}\right) = 4$$

7. *Example:* Using Geometric Formulas to Calculate Definite Integrals.

(a) Use the formula for the area of a circle to evaluate $\int_3^6 \sqrt{9 - (x - 3)^2} dx$.

(b) Find the area between the curve of the function $f(x) = 2x - 1$ and the x-axis over the interval $[-1, 3]$.

9. *Definition:* Net Signed Area

Let $f(x)$ be an integrable function defined on an interval $[a, b]$. Let A_1 be the area between $f(x)$ above the x-axis and the x-axis, and let A_2 be the area between $f(x)$ below the x-axis and the x-axis. Then

(a) the net signed area between $f(x)$ and x axis is given by $\int_a^b f(x)dx =$

(b) the total area between $f(x)$ and x axis is given by $\int_a^b |f(x)|dx =$

10. *Example.* Find the area between the curve of the function $f(x) = |2x - 1|$ and the x-axis over the interval $[0, 3]$.

11. Properties of the Definite Integral.

(a) If the limits of integration are the same, the integral is just a line and contains no area: $\int_a^a f(x)dx =$

(b) If the limits are reversed, then place a negative sign in front of the integral: $\int_b^a f(x)dx =$

(c) The integral of a sum is the sum of the integrals: $\int_a^b [f(x) + g(x)]dx =$

(d) The integral of a difference is the difference of the integrals: $\int_a^b [f(x) - g(x)]dx =$

(e) The integral of the product of a constant and a function is equal to the constant multiplied by the integral of the function.

$$\int_a^b [f(x) + g(x)]dx =$$

(f) $\int_a^b f(x)dx =$

12. Use the properties of the definite integral to express the definite integral of $f(x) = 6x^3 - 4x^2 + 2x - 3$ over the interval $[1, 3]$ as the sum of four definite integrals.

13. If it is known that $\int_1^5 f(x)dx = -3$ and $\int_2^5 f(x)dx = 4$, find the value of $\int_1^2 f(x)dx$.